Qt for HarmonyOS - Building from Source

On this page

Note: Qt for HarmonyOS is in technology preview and may change in future releases.

This page describes how to configure and build **Qt for HarmonyOS** from source.

Prerequisites

Before continuing, make sure you meet all requirements from **Qt for HarmonyOS Prerequisites**.

Building Qt for HarmonyOS is a cross-compilation: you build the Qt libraries for the HarmonyOS target using build tools from a matching host Qt build. You therefore need a Qt build for your host platform first. For general background on cross-compiling Qt, see **Qt for Device Creation** and **Cross-compiling Qt**.

Configuring and building

Configure Qt for HarmonyOS by pointing the build at the OpenHarmony SDK and the host Qt build:

```
configure -ohos-sdk <SDK_ROOT> -qt-host-path <HOST_QT_PATH> -ohos-arch arm64-v8a
cmake --build .
```

The relevant configure options are:

- `-ohos-sdk` sets the OpenHarmony SDK root path. If omitted, `configure` uses the value of the `OHOS_SDK_ROOT` environment variable.
- `-ohos-arch` sets the target architecture. Use `arm64-v8a` for physical devices and `x86_64` for the emulator. If omitted, `configure` selects an architecture based on the SDK.
- `-qt-host-path` specifies the path to the Qt host build used for cross-compiling.

Installing

After the build finishes, install Qt for HarmonyOS:

```
cmake --install .
```

By default, Qt is installed into `/usr/local/Qt-<version>`. Pass the `-prefix` option to `configure` to install into a different location. The resulting Qt for HarmonyOS installation is what you use to build and deploy applications, as described in **Building and Deploying Qt Applications for HarmonyOS**.

‹ Prerequisites

Building and deploying applications ›

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